

Project 1.5: From the Normal Distribution to χ^2 , t , and F

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All simulations use $M = 5,000$ Monte Carlo replications and a fixed random seed. Each figure shows a density-normalized histogram of the simulated values with the theoretical density overlaid in red.

Part 1: Simulating χ^2 , t , and F from normal building blocks

1.1 Chi-square ($n = 15$)

$W = \sum_{i=1}^{15} Z_i^2$ with $Z_i \stackrel{iid}{\sim} N(0, 1)$.

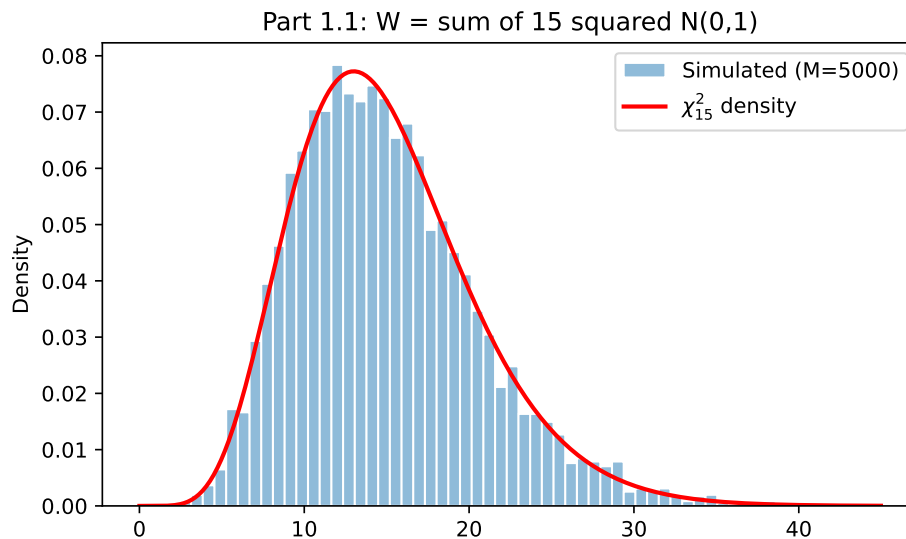


Figure 1: Simulated W vs. the χ^2_{15} density.

Comment: The histogram follows the χ^2_{15} density closely: both are right-skewed with a peak near 13 and a long right tail. The simulated mean and variance (15.00 and 31.3) are close to the theoretical values $E(W) = n = 15$ and $\text{Var}(W) = 2n = 30$.

1.2 t distribution ($n = 15$)

$T = Z/\sqrt{V/15}$ with $Z \sim N(0, 1)$ independent of $V \sim \chi^2_{15}$.

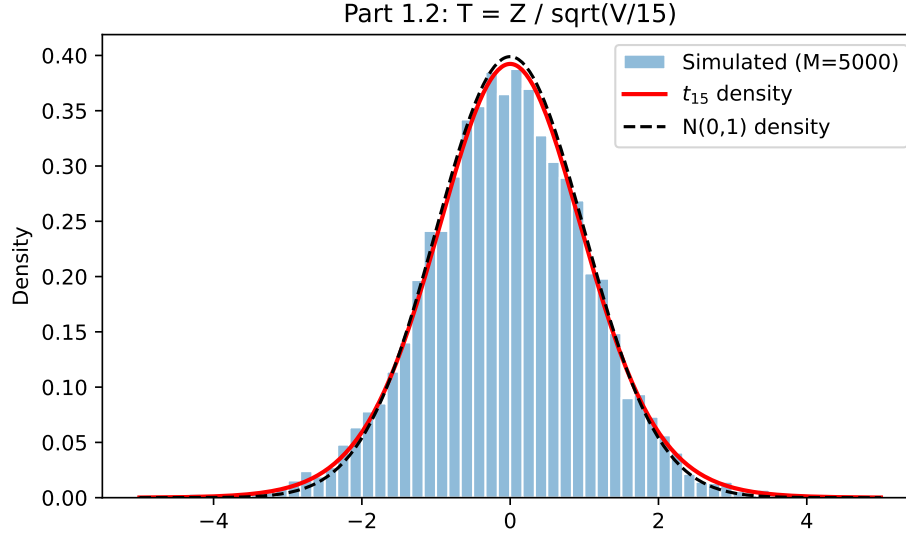


Figure 2: Simulated T vs. the t_{15} density, with the $N(0,1)$ density (dashed) for comparison.

Comment: The histogram matches the t_{15} density well. Compared with $N(0,1)$, the t_{15} curve has a slightly lower peak and slightly heavier tails, because dividing by the random quantity $\sqrt{V/15}$ adds extra variability. The difference is small at 15 degrees of freedom but shows up in the tails: $|T| > 3$ occurred in 0.92% of the simulated values, close to the t_{15} probability of 0.90% and more than three times the normal probability of 0.27%.

1.3 F distribution ($n_1 = 8$, $n_2 = 20$)

$F = (U/8)/(V/20)$ with $U \sim \chi_8^2$ independent of $V \sim \chi_{20}^2$.

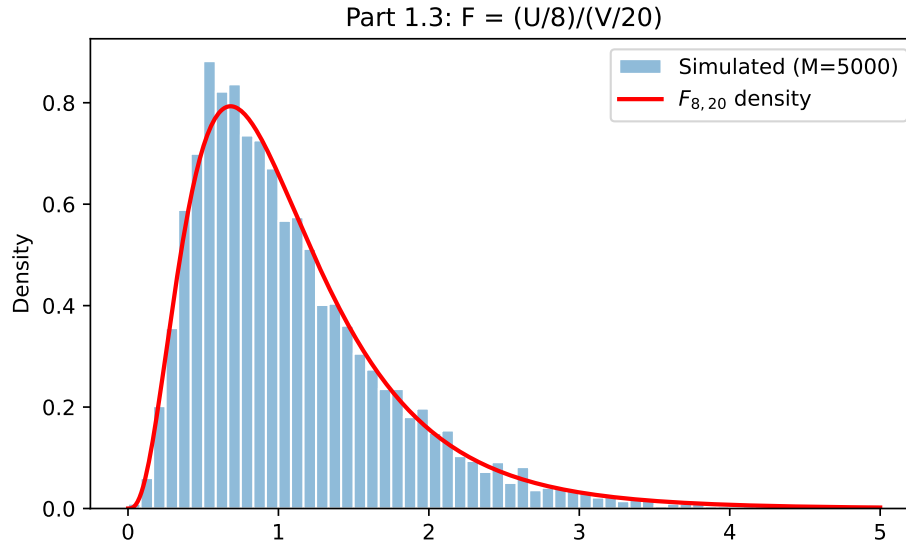


Figure 3: Simulated F vs. the $F_{8,20}$ density.

Comment: The histogram follows the $F_{8,20}$ density: it is strictly positive, peaks below 1 (around

0.7), and has a long right tail. The simulated mean of 1.10 is close to the theoretical mean $n_2/(n_2 - 2) = 20/18 \approx 1.11$.

Part 2: What happens with a real sample

$n = 20$, $\mu = 5$, $\sigma = 2$. Each repetition draws one sample $X_1, \dots, X_n \stackrel{iid}{\sim} N(\mu, \sigma^2)$ and computes

$$W' = \frac{(n-1)S^2}{\sigma^2}, \quad T' = \frac{\sqrt{n}(\bar{X} - \mu)}{S}.$$

W' vs. χ_{n-1}^2

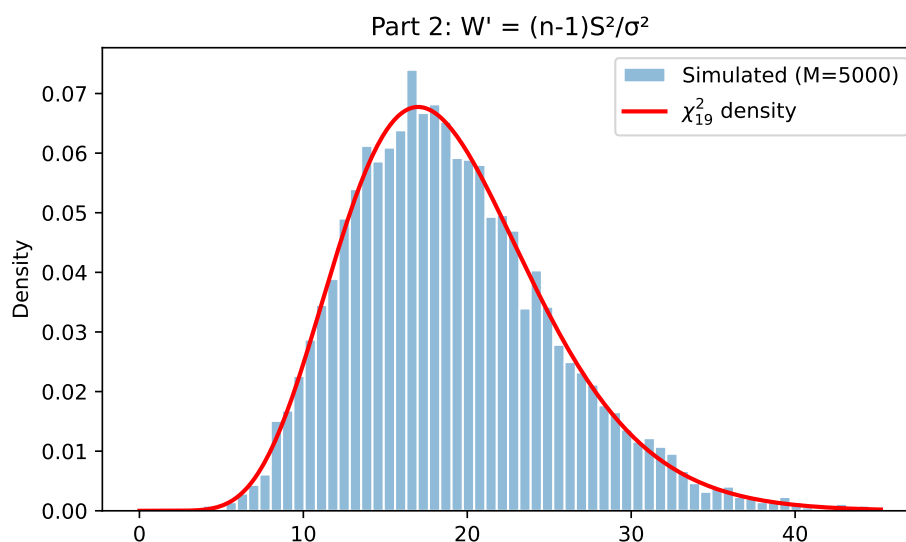


Figure 4: Simulated W' vs. the χ_{19}^2 density.

Comment: W' fits the χ_{19}^2 density closely and has the same right-skewed shape as the Part 1.1 plot, shifted right because it has more degrees of freedom. The key difference is the relationship to sample size: in Part 1.1, summing $n = 15$ squared normals gave χ_{15}^2 (degrees of freedom = n), while here a sample of $n = 20$ gives χ_{19}^2 (degrees of freedom = $n - 1$). The simulated mean of 19.05 matches 19, not 20.

T' vs. t_{n-1}

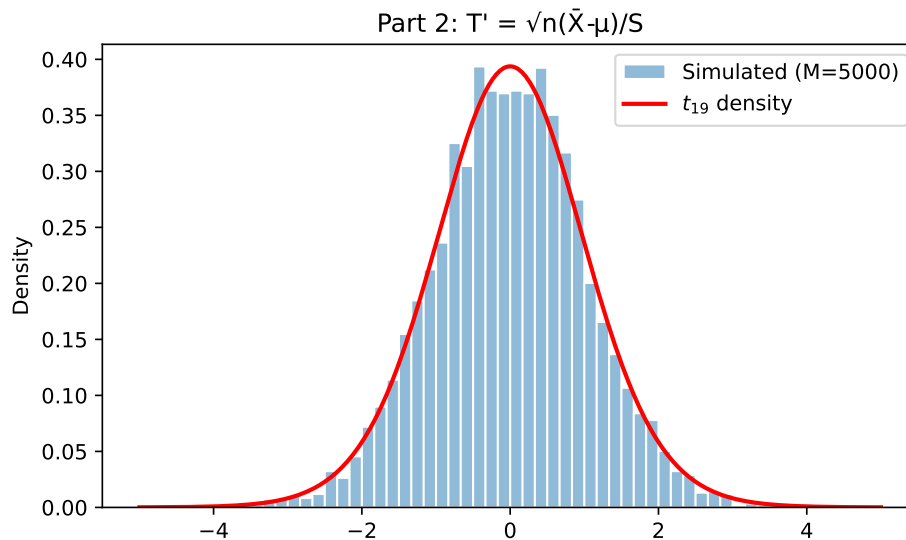


Figure 5: Simulated T' vs. the t_{19} density.

Comment: T' fits the t_{19} density well and looks like the Part 1.2 plot: symmetric, bell-shaped, and centered at 0. Even though T' was computed from a single sample with the unknown σ replaced by S , it behaves exactly like a t variable built from separate normals, with $n - 1 = 19$ degrees of freedom. With more degrees of freedom than in Part 1.2, it is even closer to $N(0, 1)$.

Why $n - 1$ degrees of freedom?

Comment: S^2 is built from the deviations $X_i - \bar{X}$, and $\bar{X}$ is computed from the same data, so these deviations must satisfy $\sum_{i=1}^n (X_i - \bar{X}) = 0$. Once any $n - 1$ of them are known, the last one is determined, so only $n - 1$ of the deviations are free to vary. Estimating $\bar{X}$ therefore uses up one degree of freedom, and W' has $n - 1$ degrees of freedom instead of n .

Part 3.1: Two samples, F_{n_1-1, n_2-1}

$n_1 = 16$, $n_2 = 21$, $\mu_1 = 3$, $\mu_2 = -2$, $\sigma_1 = 2$, $\sigma_2 = 3$.

$$F' = \frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2}.$$

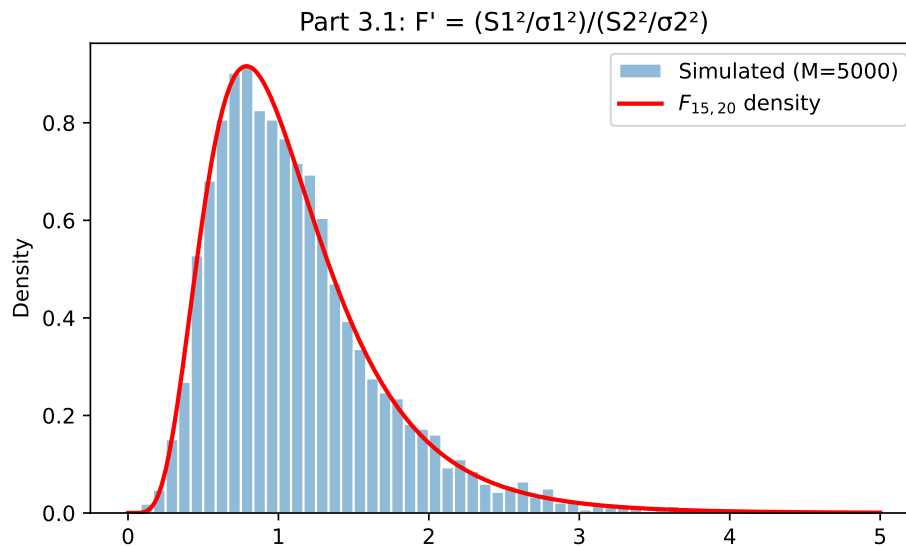


Figure 6: Simulated F' vs. the $F_{15,20}$ density.

Comment: F' fits the $F_{15,20}$ density well and has the same positive, right-skewed shape as the Part 1.3 plot. With more numerator degrees of freedom (15 instead of 8), it is more concentrated, with a taller peak closer to 1. The simulated mean of 1.12 is close to $20/18 \approx 1.11$. The different means $\mu_1 = 3$ and $\mu_2 = -2$ have no effect, because sample variances do not depend on where the data are centered.

Why $n_1 - 1$ and $n_2 - 1$?

Comment: Each sample variance is computed around its own sample mean, so each sample loses one degree of freedom: $(n_1 - 1)S_1^2/\sigma_1^2 \sim \chi_{n_1-1}^2$ and $(n_2 - 1)S_2^2/\sigma_2^2 \sim \chi_{n_2-1}^2$. Their ratio, each divided by its degrees of freedom, is therefore F_{n_1-1, n_2-1} .